

HCL Technology

Greater Noida

**Data-Driven Career Path
Recommendation Analysis**



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Team Members and Responsibilities

Team Member	Roll No./Enrollment No.	Role	Responsibilities
Divyanshu Singh	2301331530071	ML Engineer	End-to-end ML pipeline including data cleaning, data preprocessing, exploratory data analysis (EDA), feature engineering, model training, hyperparameter tuning, evaluation, and SHAP-based model explainability.
Dhruv Paliwal	2301331530068	Frontend & Deployment Engineer	Streamlit UI development, integration of ML model with frontend, and application deployment.
Surbhi Singh	2301331540161	Data & Research Analyst	Data collection, exploratory research, dataset understanding, EDA support, and report documentation.

INTRODUCTION

Problem Statement

Choosing a career path is a complex decision influenced by personality traits, academic background, and technical skills, but lack of a data-driven approach often leads to mismatched career choices and low job satisfaction.

Business Context

Industry: EdTech, Human Resources, and Career Counselling.

The Real-World Problem: Traditional counselling uses isolated data (academic or personality), failing to connect RIASEC traits with educational background, resulting in suboptimal recommendations for students, career changers, and HR professionals across EdTech and recruitment domains.

Target Users: High school students, college students, career changers, and HR recruitment professionals.

Why This Solution Matters: The system provides objective and explainable career recommendations using SHAP, reducing uncertainty in career planning and helping users identify key skills that can improve their success probability.

Objectives

Business Objective: Deliver a career analytics engine that provides personalized recommendations and predicts success probability to enable informed educational investments.

Analytical Objective: Evaluate the impact of RIASEC personality traits, educational background, and professional skills on career selection. Use SHAP to explain recommendation factors and incorporate career hierarchy mapping for career guidance.

ML Objective

- **Multi-class Classification:** Classify users into suitable career paths based on their profile.
- **Predictive Accuracy:** Achieve high prediction accuracy using the Random Forest model.
- **Feature Prioritization:** Prioritize educational qualifications and certifications through weighted feature engineering.
- **Probability Estimation:** Generate suitability and success probability scores for each recommendation.
- **Explainability (XAI):** Use SHAP to explain the key factors influencing career predictions.

Scope of the Project

- ❖ **Included:** Integration of RIASEC personality traits with academic and skill-based data, career hierarchy mapping, feature scaling, and SHAP-based Explainable AI.
- ❖ **Excluded:** Real-time job market salary data, LinkedIn profile scraping, and automated resume generation.

Data Understanding

Data Sources

The project utilizes a hybrid dataset combined from two primary sources:

1. **Career Assessment Dataset:** Contains academic scores, skill-based attributes, RIASEC personality scores, and career labels. The original dataset was synthetically extended to reach approximately 5,000 records for model training

Source: [Student Career Prediction using RIASEC Dataset](#)

2. **Education & Skills Dataset:** Contains educational qualifications, specializations, skills, certifications, CGPA, and recommended career paths.

Source: [Career Path Recommendations Dataset](#)

Dataset Description

Dataset 1: Career Assessment Dataset

Column Name	Data Type	Description	Typical Range
Math_Score	Numeric	Mathematics performance score	0-100
Science_Score	Numeric	Science performance score	0-100
R_score	Numeric	Realistic personality score	0-50
I_score	Numeric	Investigative personality score	0-50
A_score	Numeric	Artistic personality score	0-50
S_score	Numeric	Social personality score	0-50
E_score	Numeric	Enterprising personality score	0-50
C_score	Numeric	Conventional personality score	0-50
Career	Categorical	Target career category	Multiple Classes

Dataset 2: Education & Skills Dataset

Column Name	Data Type	Description	Typical Range
Education Level	Categorical	Highest qualification	High School, UG, PG
Specialization	Categorical	Academic specialization	Various
Skills	Text	Technical and professional skills	Multiple Values
Certifications	Text	Certifications completed	Various
CGPA/Percentage	Numeric	Academic performance indicator	0-10 / 0-100
Recommended Career	Categorical	Suggested career path	Multiple Classes

Initial Data Exploration

- **Missing Values and Fixing:**
 - Certifications in the Education dataset had 596 missing values, which were filled with "None".
 - Used dropna() before modeling to ensure target alignment.
- **Outliers and Fixing:**
 - Boxplots were generated (Phase 3) to identify variance in RIASEC scores. Standard Scaling was used to handle the magnitude differences.
- **Class Distribution:**
 - The target variable Unified_Career was analyzed and found to be balanced after the merge, with ~400 samples per class.
- **Encoding:**
 - MultiLabelBinarizer was used to expand the 'Skills' column into individual binary features.
 - LabelEncoder was applied to Education Level, Specialization, and Certifications.
- **Feature Engineering:**
 - **Career Mapping:** Created a Unified_Career superset to merge labels from different sources.
 - **Education Boost:** Applied a 5.0x weight factor to academic features to prioritize qualifications in the recommendation logic.
 - **Career Hierarchy:** Developed a progression graph to map entry-level roles to senior leadership positions.
- **Other Data Quality Issues:**
 - Detected and removed 2,600 duplicate rows in the RIASEC dataset and 1 duplicate in the Education dataset to prevent model overfitting.

Model Development

Model Selection

1. **Logistic Regression:** Chosen as a linear baseline. While it performed well (~95.21%), it initially struggled with convergence, indicating that the relationships between personality scores and careers may be non-linear.
2. **Decision Tree:** Chosen for its interpretability and ability to handle both numerical and categorical features. It achieved an accuracy of **94.38%** and provided clear decision-making logic, but its performance was slightly lower than the other models and it was more prone to overfitting.
3. **Random Forest (Final Model):** Achieved the highest accuracy (**95.21%**) and was selected as the final model. It effectively handles the high dimensionality of the **merged dataset (16+ features)**, captures complex patterns, reduces overfitting, and provides better generalization than a single decision tree, making it the most reliable model for career recommendation.

Model Training

- **Process:** The data was split into an **80/20 train-test ratio** using stratification to maintain class balance across all 6 career categories.
- **Initial Parameters:** Used default settings with `random_state=42` to ensure reproducibility.
- **Education Weighting:** To prioritize real-world eligibility, a custom **5.0x boost factor** was applied to the Education Level, Specialization, Certifications, and CGPA features during the final hybrid training phase.

Hyperparameter Tuning

To maximize performance, RandomizedSearchCV was implemented (Phase 9) with 3-fold cross-validation.

- **Why it was done:** Tuning was necessary to find the optimal balance between model depth and the number of trees, preventing the model from becoming too complex and memorizing the training data (overfitting).
- **Best Parameters Found:**
 - **n_estimators:** 300 (Increased Forest size for stability).
 - **max_depth:** 20 (Limited depth to prevent extreme specialization).
 - **min_samples_leaf:** 4 (Ensures each leaf has enough data points to be statistically significant).
- **Result:** The tuning process successfully increased the test accuracy from **95.21%** to **95.42%**.

Model Evaluation

The hybrid career recommendation model was evaluated on a 20% held-out test set. The final tuned **Random Forest** model achieved an overall accuracy of **95.42%**.

Classification Performance

The model performs exceptionally well across all categories, with expected, minor overlaps in closely related tech fields.

Career Category	Precision	Recall	F1-Score
Accounting & Finance	1.00	1.00	1.00
Business & Marketing	1.00	1.00	1.00
Data Science & AI	0.83	0.90	0.86
Education & Counselling	1.00	1.00	1.00
Healthcare	1.00	1.00	1.00
Software Development	0.89	0.81	0.85
Weighted Average	0.95	0.95	0.95

Confusion Matrix Analysis

The confusion matrix showed near-perfect classification performance. Most career categories were classified correctly, with only minor confusion observed between Data Science & AI and Software Development due to overlapping technical skill requirements.

Cross-Validation & Stability

To guarantee generalization, **3-fold Cross-Validation** was embedded within the RandomizedSearchCV hyperparameter tuning process (evaluating 10 candidate combinations). Consistent performance across folds confirms the model is highly stable and robust against data splitting bias.

Feature Importance (Explainable AI)

Using SHAP (SHapley Additive exPlanations), we mapped the exact drivers behind the model's decisions:

- Global Importance:** Psychological traits dictate the baseline direction. For example, high E_score (Enterprising) drives Business roles, while I_score (Investigative) anchors Research paths.
- Local Explanations:** Individual patient/student audits show that while personality sets the initial trajectory, the **Education Boost Factor** (CGPA and specific Certifications) acts as the final validator for the output.
- Directional Impact:** SHAP force plots reveal logical decision boundaries—possessing a skill like *Python* pushes predictions toward Data Science while actively penalizing Healthcare suitability.

Results and Findings

Model Performance Summary

The development process concluded with a high-performing hybrid recommendation engine. The performance metrics across different stages were as follows:

- **Baseline Performance:** The initial Random Forest and Logistic Regression models achieved **95.21%** accuracy.
- **Final Optimized Performance:** After applying RandomizedSearchCV for hyperparameter tuning, the accuracy improved to **95.42%**.
- **Category Reliability:** The model demonstrated 100% precision in several critical fields, including Healthcare and Accounting & Finance, proving it is highly reliable for specialized career paths.
- **Inference Speed:** The model is optimized for real-time deployment, as demonstrated by the 'Enhanced Career Prediction Dashboard' which generates predictions and SHAP visualizations in seconds.

Key Findings

1. **Personality vs. Education:** While RIASEC personality scores (specifically Investigative and Social) are strong indicators of career *interest*, academic background (Specialization and CGPA) acts as the primary *eligibility* filter.
2. **The Impact of Weighting:** Applying a **5.0x boost** to education-related features significantly reduced 'aspirational' errors (where a user has the personality but not the qualifications) without sacrificing overall model accuracy.
3. **Skill Synergy:** The SHAP analysis revealed that specific technical skills like 'Programming' and 'Logical Ability' have a strong synergistic effect with the Investigative (I) personality score, making them the top factors for 'Data Science & AI' roles.
4. **Career Progression Logic:** Instead of suggesting an entry-level role in isolation, the dashboard provides a roadmap from entry-level (e.g., Sales Assistant) to executive leadership (e.g., CEO), ensuring the recommendation remains relevant for the user's entire career journey.
5. **Explainability:** The integration of SHAP force plots successfully transitioned the model from a 'black box' to an interactive counselling tool, providing users with the 'Why' behind every career recommendation.

Deployment

Tools Used

- **Python** – Core programming language
- **Streamlit** – Web application framework for building the user interface
- **Scikit-learn** – Machine learning model development
- **Pandas & NumPy** – Data preprocessing and analysis
- **SHAP** – Explainable AI and model interpretation
- **Matplotlib** – Data visualization
- **Joblib** – Model serialization and loading
- **Git & GitHub** – Version control and source code hosting
- **Streamlit Community Cloud** – Application deployment and hosting

GitHub Repository - <https://github.com/DivyanshuSingh-12/Career-recommendation-System>

Streamlit link - <https://career-recommendation-system-hcl.streamlit.app/>

Challenges Faced

1. Dataset Integration

Two different datasets (psychometric/RIASEC and education/career data) were merged, requiring alignment of formats and feature consistency.

2. Data Preprocessing

Handling missing values, inconsistent labels, and unstructured skill data required extensive cleaning and standardization.

3. Feature Engineering (16+ features)

The merged dataset increased complexity, requiring proper encoding and transformation of numerical and categorical variables.

4. Encoding Categorical Data

Features like Education Level, Specialization, Certifications, and Skills were encoded using label encoding and multi-label techniques.

5. Model Selection

Multiple models were evaluated, and Random Forest was selected for its higher accuracy and ability to handle non-linear relationships.

6. Explainability (SHAP)

Integrating SHAP for multi-class predictions required careful extraction and interpretation of feature contributions.

7. Streamlit Deployment

Managing session state, UI interactions, and real-time prediction flow required debugging and optimization.

8. Git & Version Control

Resolving commit conflicts and non-fast-forward push issues required proper Git workflow management.

Future Enhancements

1. **LinkedIn Integration (Professional Profile Mapping)**

Integrating LinkedIn profiles to analyse real-world user data such as experience, skills, certifications, and internships for more accurate and industry-aligned career recommendations.

2. **Resume & Skill Gap Analysis**

Adding an intelligent system that evaluates user resumes against industry job descriptions to identify missing skills and suggest personalized improvements like courses, certifications, and projects.

3. **Real-Time Job Market Integration**

Connecting with live job platforms (e.g., LinkedIn Jobs, Indeed, Naukri) to recommend careers based on current hiring trends, job availability, and in-demand skills.

4. **Salary Prediction Module**

Introducing a predictive model to estimate salary ranges for recommended careers based on skills, experience level, job role, and market conditions.

5. **Enhanced Explainability System**

Improving SHAP-based explanations with interactive and user-friendly visual dashboards to clearly show why a specific career was recommended.

6. **End-to-End Career Ecosystem Platform**

Evolving the system into a complete career guidance platform that combines career prediction, job matching, resume evaluation, and skill development in a single integrated solution.

Conclusion

The Career Recommendation System successfully demonstrates how machine learning can be applied to guide students and professionals in making informed career decisions. By integrating psychometric scores, RIASEC personality traits, and educational background, the system provides personalized career predictions using a trained Random Forest model.

The model achieved strong performance and effectively handled a merged dataset of 16+ features, capturing complex relationships between skills, interests, and career outcomes. Additionally, the inclusion of SHAP explainability improves transparency by showing the key factors influencing each prediction.

The Streamlit-based application provides an interactive and user-friendly interface, allowing users to input their details, view career recommendations, explore alternative options, and download detailed reports.

Overall, this project bridges the gap between data-driven analytics and real-world career guidance, making career planning more structured, transparent, and accessible.

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